



Database Management Systems

Correlated Nested Query

Anchor:

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Database Management Systems

Unit 2: Advanced SQL

Slides adapted from Author Slides of “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

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- A **correlated nested query** can be defined as a nested query in which the WHERE clause references **attributes of a relation declared in the outer query**.
- A correlated nested query is **evaluated once for each tuple** (or combination of tuples) of the relation in the outer query.

Example: Retrieve the name of each employee who has a dependent who is of the same gender as the employee.

Query: SELECT E.Fname, E.Lname FROM EMPLOYEE AS E WHERE E.Ssn IN
(SELECT D.Essn FROM DEPENDENT AS D WHERE E.Gender = D.Gender);

Output:

Fname	Lname
Franklin	Wong

- For each *EMPLOYEE* tuple, the nested query is evaluated, and the *Essn* values of all *DEPENDENT* tuples with the same *Gender* as that of the *EMPLOYEE* tuple are retrieved.
- If the *Ssn* value of the *EMPLOYEE* tuple is in the result of the nested query, the *EMPLOYEE* tuple is selected
- In general, a query written with nested SELECT-FROM-WHERE blocks and using the = or IN comparison operators can always be expressed as a single block query.
- For the current example, the equivalent query without a nested query is given below:

```
SELECT DISTINCT E.Fname, E.Lname  
FROM EMPLOYEE AS E, DEPENDENT AS D  
WHERE E.Ssn = D.Essn AND E.Gender = D.Gender;
```

- **EXISTS** and **UNIQUE** are Boolean functions that can be used in the WHERE clause
- The EXISTS function in SQL is used to check whether the **result of a nested query is empty** (contains no tuples) or not.
- **The result of EXISTS is True if the nested query result contains at least one tuple, and False if the nested query result contains no tuples.**
- On the other hand, NOT EXISTS returns True if there are no tuples in the result of the nested query, and False if the nested query result contains at least one tuple.
- EXISTS and NOT EXISTS are typically used in conjunction with a correlated nested query.

Example: Retrieve the name of each employee who has a dependent who is of the same gender as the employee.

Query: SELECT E.Fname, E.Lname FROM EMPLOYEE AS E
WHERE EXISTS (SELECT * FROM DEPENDENT AS D
WHERE E.Ssn = D.Essn AND E.Gender = D.Gender);

Output:

Fname	Lname
Franklin	Wong

- For each *EMPLOYEE* tuple, the nested query is evaluated and it returns all *DEPENDENT* tuples with the same *Essn* and *Gender* as the *EMPLOYEE* tuple.
- If at least one tuple exists in the result of the nested query, then the *EMPLOYEE* tuple under consideration is selected.

Example: Retrieve the names of employees who have no dependents

Query: SELECT Fname, Lname FROM EMPLOYEE

WHERE NOT EXISTS (SELECT * FROM DEPENDENT WHERE Ssn = Essn);

Output:

Fname	Lname
Joyce	English
Ramesh	Narayan
James	Borg
Ahmed	Jabbar
Alicia	Zelaya

- For each *EMPLOYEE* tuple, the correlated nested query selects all *DEPENDENT* tuples whose *Essn* value matches the *Ssn* value in the *EMPLOYEE* tuple
- If the result is empty (i.e. no dependents), the *EMPLOYEE* tuple is selected as the condition in the WHERE clause evaluates to True.

Example: List the names of managers who have at least one dependent.

Query: SELECT Fname, Lname FROM EMPLOYEE
WHERE EXISTS (SELECT * FROM DEPENDENT WHERE Ssn = Essn)
AND EXISTS (SELECT * FROM DEPARTMENT WHERE Ssn = Mgr_ssn);

Output:

Fname	Lname
Franklin	Wong
Jennifer	Wallace

- For each *EMPLOYEE* tuple, the first nested query selects all *DEPENDENT* tuples related to the *EMPLOYEE* tuple.
- The second nested query selects all *DEPARTMENT* tuples managed by *EMPLOYEE* tuple.
- The *EMPLOYEE* tuple is selected if both the nested queries return at least one tuple each.

Example: Retrieve the name of each employee who works on all the projects controlled by department number 4

There are two ways to write this query

Query: SELECT Fname, Lname FROM EMPLOYEE

WHERE NOT EXISTS ((SELECT Pnumber FROM PROJECT WHERE Dnum = 4)

EXCEPT (SELECT Pno FROM WORKS_ON WHERE Ssn = Essn));

Output:

Fname	Lname
Ahmed	Jabbar
Alicia	Zelaya

(SELECT Pnumber FROM PROJECT WHERE Dnum = 4)

- This nested query retrieves the project number of all projects managed by department 4. It is not correlated with the outer query.

(SELECT Pno FROM WORKS_ON WHERE Ssn = Essn)

- This nested query is correlated with the outer query. For each *EMPLOYEE* tuple, it retrieves the project number of all projects that the employee works on.

((SELECT Pnumber FROM PROJECT WHERE Dnum = 4)

EXCEPT (SELECT Pno FROM WORKS_ON WHERE Ssn = Essn));

- Set difference operation is performed between the results of the nested queries i.e. set of project numbers under department 4 MINUS set of project numbers that a particular employee works on.
- The result is the set of project numbers controlled by department 4 that the employee under consideration does not work on.
- If that particular employee works on all projects controlled by department 4, the result of the above query would be an empty relation.
- The condition in the WHERE clause would be satisfied and the *EMPLOYEE* tuple under consideration would be selected

The second way to write this query

Query: SELECT Fname, Lname FROM EMPLOYEE E
WHERE NOT EXISTS (SELECT * FROM WORKS_ON B
WHERE (B.Pno IN (SELECT Pnumber FROM PROJECT
WHERE Dnum = 4)
AND NOT EXISTS (SELECT * FROM WORKS_ON C
WHERE C.Essn = Ssn AND C.Pno = B.Pno)));

Output:

Fname	Lname
Ahmed	Jabbar
Alicia	Zelaya

```
SELECT Fname, Lname FROM EMPLOYEE
WHERE NOT EXISTS ( SELECT * FROM WORKS_ON B
                    WHERE ( B.Pno IN ( SELECT Pnumber FROM PROJECT WHERE Dnum = 4 )
                    AND NOT EXISTS ( SELECT * FROM WORKS_ON C
                                    WHERE C.Essn = Ssn AND C.Pno = B.Pno )));
```

Each color represents a level of nesting

Let us try to understand how this query works:

- There is an **outer query** with 2 levels of nested subqueries. The **outermost nested query** is correlated with the **outer query**. There are **two inner subqueries** nested in the **outermost nested subqueries** – one which is correlated with the **outermost nested query** and the other which is not.

- The **outermost nested query** is evaluated for each *EMPLOYEE* tuple.
- As mentioned, one of the **inner nested queries** is not correlated with the **outermost nested query**:

(SELECT Pnumber FROM PROJECT WHERE Dnum = 4)

- This query returns the project number of all projects controlled by department 4.
- The other inner nested query is correlated and is therefore evaluated for every *WORKS_ON* tuple in the **outermost nested query**:

(SELECT * FROM WORKS_ON C WHERE C.Essn = Ssn AND C.Pno = B.Pno)

- For each *EMPLOYEE* tuple and *WORKS_ON* tuple, this query selects all tuples from the *WORKS_ON* relation whose *Essn* value matches the *Ssn* of the *EMPLOYEE* tuple and project number matches the *Pno* of the outer *WORKS_ON* tuple.

- Now, let's look at the **outermost nested query** (evaluated for every *EMPLOYEE* tuple) :

```
( SELECT * FROM WORKS_ON B WHERE ( B.Pno IN  
( SELECT Pnumber FROM PROJECT WHERE Dnum = 4 )  
AND NOT EXISTS ( SELECT * FROM WORKS_ON C  
WHERE C.Essn = Ssn AND C.Pno = B.Pno )));
```

- This query selects only those *WORKS_ON* tuples that are concerned with projects controlled by department 4 and which the employee under consideration does not work on.
- If the employee under consideration works on all projects under department 4, then there would be no tuples selected by the **outermost nested query**. This would evaluate the WHERE predicate to True (as we are using NOT EXISTS) and hence, the *EMPLOYEE* tuple under consideration would be selected

1) What distinguishes a correlated nested query from a standard (uncorrelated) nested query?

- A) It cannot use the EXISTS operator.
- B) Its WHERE clause references attributes of a relation declared in the outer query.
- C) It executes only once for the entire query statement.
- D) It must always be placed within the FROM clause.

2) Fill in the blank: Unlike a standard subquery, a correlated nested query is evaluated once for each _____ of the relation in the outer query.

3) When using Boolean functions in SQL, what does the EXISTS function evaluate to if its nested query result contains no tuples?

- A) TRUE
- B) UNKNOWN
- C) FALSE
- D) NULL

4) The Boolean functions EXISTS and NOT EXISTS check whether the result of a subquery is empty or not, and they are typically used in conjunction with a _____ nested query.

5) When utilizing the EXCEPT operator to perform a set difference between two nested queries, what happens if the first set is completely subtracted by the second set (e.g., an employee works on all the required projects)?

- A) It returns a NULL value.
- B) It throws a syntax error.
- C) It returns all tuples from the first set.
- D) It results in an empty relation.

Answers:

1. B
2. Tuple
3. C
4. Correlated
5. D



THANK YOU

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